

Bare Metal Server Install

GNS3

General Information:

- This guide was developed and tested with Ubuntu Server 18.04 (LTS), however I personally use this on my production systems running Ubuntu Server 19.10 with no issues
- Text in this document following the # symbol denotes comments or additional instruction
- Text surrounded in [] denotes keys to press on your keyboard
- `Text in this format` is generally commands that will need to be typed in, or the output of a command for comparison
- This guide will **NOT** cover the installation method for Ubuntu Server, however there are a couple of things I recommend doing during the install:
- Set a static IP Address during the install (write it down and/or make a note of it somewhere handy, you will need it later)
- During the install it will ask if you want to install the openssh server, I highly recommend you do so. This will allow you to remotely manage the server later and during the GNS3 Server install process. If you did install it during the install of the operating system, you can simply SSH from another system to the server and execute this guide via SSH. If you did not install openssh server during the Ubuntu install, here is the command to do it after words:

```
sudo apt-get update && sudo apt-get install openssh-server
```

- This guide will also **NOT** cover copying existing projects or appliances from an old server/VM over to the new one
- REMEMBER: Linux is case sensitive!

Updating the operating system and software on the Ubuntu Server:

Following a fresh install of Ubuntu, it is highly recommended that you update the operating system and software

```
sudo apt-get update && sudo apt full-upgrade
```

This updates the cache on the system with the Ubuntu repositories, then downloads and upgrades your system

Once the update is complete, you should reboot the system in order to boot off the new kernel and various other software that may have been updated

```
sudo reboot
```

Checking to ensure that virtualization is enabled and supported on the system:

First we need to install a small piece of software in order to check for virtualization support

```
sudo apt-get install cpu-checker
```

Once it is installed we can run the below command

```
sudo kvm-ok
```

This should result in the following output if virtualization is supported

```
INFO: /dev/kvm exists
KVM acceleration can be used
```

Installing the GNS3 Server Software:

First we need to add the repository (PPA) that hosts the GNS3 software

```
sudo add-apt-repository ppa:gns3/ppa
```

Next we will update the repository caches again

```
sudo apt-get update
```

Now we actually download and install the GNS3 Server software

```
sudo apt-get install gns3-server
```

Installing Docker support:

If you want to be able to run Docker based appliances on your server, you will also need to install docker.io

```
sudo apt-get install docker.io
```

Configuring the GNS3 Server software to start at boot (running as a service):

To enable the server software to run as a service, we will need to create and edit a text file. The instructions below are for using vi, which is what I normally use for text editing in a Linux CLI. VIM is another option and is a bit more friendly to people not used to working on the command line.

First we will open the file in vi so that we can edit it. The file does not exist yet, but will be written to the hard drive once we are done

```
sudo vi /etc/systemd/system/gns3.service
```

Now we will enter into "append mode" in vi in order to add the text we need

```
Press the [a] key on your keyboard
```

If you are connected to the server via an ssh connection, you can simply copy and paste the following text in at this point. If not, and you are doing this locally on the server, you will need to type the following out

```
[Unit]
Description=GNS3 Server

[Service]
ExecStart=/usr/share/gns3/gns3-server/bin/gns3server

[Install]
WantedBy=multi-user.target
```

We now need to exit out of "append mode" and save the file, while also quitting vi

```
Press the [ESC] key on your keyboard twice (the second time is just to
make sure)
Press the following keys in sequence: [:] [w] [q] [!] [ENTER]
```

Now that we have created the file that has the configuration information for the service, we will now enable the service and start it for the first time

The first command makes the file we created executable

The second command creates a symbolic link to that file and enables it as a service

The final command, actually starts the service

```
sudo chmod +x /etc/systemd/system/gns3.service
sudo systemctl enable gns3
sudo systemctl start gns3
```

Now that we have started the service, let's verify that the service is running, and also validate that it will start automatically after a reboot

Check to see if the GNS3 server is running:

```
sudo systemctl status gns3
```

You should see output similar to the below, we are looking for the **BOLD** portion specifically to see that the service is active and running

```
gns3.service - GNS3 Server
   Loaded: loaded (/etc/systemd/system/gns3.service; enabled; vendor
preset: ena
   Active: active (running) since Wed 2020-01-01 23:31:35 UTC; 3min 25s
ago
 Main PID: 1140 (gns3server)
    Tasks: 1 (limit: 4915)
   Memory: 71.0M
   CGroup: /system.slice/gns3.service
           └─1140 /usr/share/gns3/gns3-server/bin/python
/usr/share/gns3/gns3-se     # The rest of the output is omitted...
```

Now let's go ahead and reboot the system to see if the service starts automatically at boot

```
sudo reboot
```

Once the server has rebooted and is up and running, login via ssh or locally on the system and run

```
sudo systemctl status gns3
```

You should see output similar to what we saw before when we checked the status. If you do, **CONGRATULATIONS,** you now have a working bare-metal GNS3 server!

Configure GNS3 Client software to use the new server:

- Start your GNS3 client software
- Edit->Preferences->Server
- Uncheck the "Enable Local Server" checkbox
- Under "Host" enter in the IP address of the bare-metal GNS3 Server
- Uncheck "Auth"
- Click "OK" button in the bottom right
- It may be necessary to close the GNS3 client software and restart for the changes to fully be applied
- At this point you should see the new server listed in the GNS3 GUI
- You will have to re-import any appliances you have into the new server

Additional software to install that may be useful:

If you want to be able to see the utilization of the servers CPU, install htop. you will be able to SSH to the server and run htop to see the CPU/Memory status

```
sudo apt-get install htop
```

if you want to monitor CPU core temperatures, install lm-sensors

```
sudo apt-get install lm-sensors
```

To get updates every second for the core temps, run the below from the command line on the server or over SSH:

```
watch -n 1 sensors
```